MATTHEW ZHOU

916-201-1280 | matthewzhou05@gmail.com | LinkedIn

EDUCATION

Georgia Institute of Technology [May 2027 (Expected)]

- **B.S. Mechanical Engineering**, Concentration in Automation & Robotics, Minor in German (GPA: 3.97/4.00).
- Relevant Coursework: Robotics, Machine Design, Design & Manufacturing, Heat Transfer, ME Systems Laboratory

EXPERIENCE

RoboRambler GT (ARCC/RoboMaster Competition Team) | Co-Founder, Ex-President & Mechanical Lead [September 2023 – Present]

Designing and manufacturing two fleets of first-person-operating mobile combat robots armed with artillery

- Finished **2nd** in 1v1 competition and 9th in 3v3 competition among over 30 teams worldwide during ARCC 2026 (formerly RoboMaster North America); debuted and finished 15th in 1v1 competition and 17th in 3v3 competition in RoboMaster NA 2025.
- Obtained workspace in Tech Square ATL in 2025 and helped start Humanoids @ GT club to diversify student robotics endeavors.
- Developed X-drive chassis with suspension, eliminating off-road FPV blur and traction loss due to aggressive maneuver, featuring custom omni-wheel design with carbon fiber plates and molded TPU roller, achieving 20% less weight than stock wheels.
- Designed gun weight compensation mechanism with trigonometry, consuming near 0 power to aim a 3-kg gun at any pitch angle.
- Used heat-treated PA6-CF prints, pocketed CF plates, EDM-machined aluminum extrusions, and CNC metal gun chambers for 2026 fleet, coordinated tolerancing, finishing, and logistics with international manufacturers, making robots 8% lighter than 2025 fleet.
- Developed 20-RPS 17mm-caliber flywheel artillery and 42mm-caliber with <0.1 m dispersion at 5 m with custom curved flywheels.
- Secured donated through-bore slip rings, enabling infinite turret rotation on chassis with reliable electrical connection and ammo supply; led talks with companies for spatial memory unit, 3D printing filament, and humanoid actuator donations worth $7,000+.

Volkswagen AG | Assembly Maintenance Engineering Intern at Chattanooga Plant [May 2026 – Present]

Providing remedy solutions to VW Atlas production lines and saving costs

- Designed mechanical-driven push skid system sweeper with helical brush, reducing faults and estimated to save 5 lost cars monthly.
- Modified glue cell robot gripper and sensor configuration to accommodate VW 2027 Atlas quarter glass, proposed making rapid 3D printed prototype to allow early robot re-program start, saving the company over $90,000 outsourcing cost and 6 weeks of lead time.
- Reverse-engineered and 3D-printed replacements for expensive proprietary plastic components on equipment; supported a finance presentation advocating investment to quadruple the plant's 3D-printing capacity.

Advanced Manufacturing Pilot Facility (GT Manufacturing Institute) | Research Assistant [January 2024 – Present]

Worked on projects that facilitate mechanical testing automation and support advanced manufacturing techniques

- Received President's Undergraduate Research Award for a study adapting twin bridge geometry to investigate anisotropic behaviors in circular metal extrusions, determined testable & machinable sample dimension based on material property, load and grip capacity.
- Created and FEA-validated fixture for the designed sample, compatible with torsional and axial load case and DIC mirror mounting.
- Strain-validated an induction-heated printing buildplate with 60 lb. print capability with hand-soldered strain gauges and FEA.
- Certified as student coach on Wire EDM with 100+ hours of experience, trained peers, and machined alloy testing samples.
- Presented facility Directed Energy Deposition air supply analysis at GT Mfg. 4.0 Consortium, showcasing scheduling strategies.

Marine Robotics Group at Georgia Tech | Mechanical Lead for RoboSub Team [September 2023 – May 2025]

Developed a fully autonomous vehicle performing maneuver and manipulation tasks underwater

- Co-designed main vehicle hull and thruster configurations that empower the vehicle with 6-DOF underwater maneuverability.
- Engineered gripper, external battery tube, torpedo launcher, and sensor mounts to accomplish underwater manipulation tasks.
- Developed a neutrally buoyant autonomous torpedo with 2 rear thrusters, capable of self-guiding to a 4-in-diameter target from 3 ft.
- Achieved team's #17 placement out of 40 at RoboSub 2025 through leadership in mechanical design and subsystem integration.

SKILLS

Software: SolidWorks (SOLIDWORKS Design Professional (CSWP)), CATIA V5, Onshape, Microsoft Office Suite.

Programming: MATLAB, Arduino, Java.

Languages: English, German, Chinese (Mandarin), Chinese (Cantonese)

Others: 3D Printing & Heat Treatment, Project Management, Corporate Sponsorship Negotiation, Soldering, Wire EDM, Mill, Lathe, FEA.